

How to treat autism in children?

Evidence searched on 2026-10-06 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Bottom line Exercise interventions show a dose-response relationship improving motor skills in children with autism spectrum disorder (ASD), with significant gains in overall, gross, and fine motor function (Wang et al., *European Journal of Pediatrics*, 2026 [PMID 42758334]).

Most important recent finding Exercise significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response relationships identified for overall and gross motor skills. (Wang et al., *European Journal of Pediatrics*, 2026 [PMID 42758334])

Immediate context Motor impairments are prevalent in children with ASD and may impact functional independence. While exercise shows promise for motor skills, **no direct evidence** from retrieved sources addresses social communication or behavioral symptoms. Ongoing trials explore interventions like occupational therapy and structured walking, but results are pending.

Top treatment drugs (from FDA drug labels):

Drug	Category	FDA-labelled indication (excerpt)
Aripiprazole	FDA-approved for the labelled indication	"1 INDICATIONS AND USAGE Aripiprazole is indicated for the treatment of: • Schizophrenia • Irritability Associated with Autistic Disorder • Treatment ..."
Risperidone	FDA-approved for the labelled indication	"with lithium or valproate, for the treatment of acute manic or mixed episodes associated with Bipolar I Disorder (1.2) Treatment of irritability ..."

Latest findings

2026

- **Behavioral sleep interventions:** A systematic review of 39 studies (2002–2026) reported **significant improvements in sleep outcomes** for autistic children under 18 using behavioral strategies (Mazurek et al., *Sleep Medicine*, 2026 [PMID 42685447]).
- **Oral care:** No new direct evidence on tailored oral hygiene interventions for autistic children was retrieved (Floríndez et al., *Autism*, 2025 [PMID 39056277]).
- **Nature-based activity:** Qualitative study (n=15 parents) reported **increased communication and calmness** in preschool autistic children during nature walks, though coordination challenges were noted (Galbraith et al., *International Journal of Qualitative Studies on Health and Well-being*, 2026 [PMID 42603141]).

Current evidence

Evidence-supported but limited

- **Motor skills:** Exercise interventions (e.g., dance, structured walking) are **evidence-supported but limited** for improving motor skills in ASD, with dose-response (exploratory) effects (Wang et al., 2026 [PMID 42758334]).

Emerging

- **Sleep:** Behavioral interventions (e.g., bedtime routines, sleep restriction) are **evidence-supported but limited** for sleep problems in autistic children (Mazurek et al., 2026 [PMID 42685447]).
- **Emotion regulation:** Non-pharmacological therapies (e.g., CBT-adapted protocols) show **promising but inconsistent** effects on emotion dysregulation (Restoy et al., 2024 [PMID 38401510]).
- **Social communication:** Music therapy dosage remains **under investigation** (Chen et al., *BMJ Open*, 2026 [PMID 42486527] [protocol]; no results retrieved).

Experimental or preliminary

- **Oral hygiene:** Peer modeling and video demonstrations are **under investigation** for improving plaque control in autistic children (NCT07859696, *not yet recruiting*).

Indirect evidence

- **Genetic/environmental overlap:** Twin studies suggest **shared genetic/environmental factors** between autistic traits and anxiety/depression, but no direct ASD treatment implications (Han et al., *Journal of Affective Disorders*, 2026 [PMID 42508582]). (*indirect evidence: different or related population*)
- **Feeding difficulties:** A structured algorithm (FEEDSETUP) addresses pediatric feeding challenges in **related populations** (Fisberg et al., *Jornal de Pediatria*, 2026 [PMID 42716484]).

Insufficient evidence

- **Nutraceuticals:** Omega-3/inositol and N-acetylcysteine showed **trends toward benefit** for depression/mania in emotionally dysregulated youth *with or without ASD* (Wozniak et al., *Journal of Affective Disorders*, 2026 [PMID 42462928]); **not specific to ASD**.

Protocol – no results

- Dance programs (AllPlay Dance; Chadha et al., *BMJ Open*, 2026 [PMID 42711081]), music therapy (Chen et al., 2026 [PMID 42486527]), and water safety (Kemp et al., 2026 [PMID 42552003]) are **under investigation**; no efficacy established.

Non-drug treatments

- **Exercise:** Structured programs improve motor skills (Wang et al., 2026 [PMID 42758334]).
- **Behavioral sleep interventions:** Effective for sleep problems (Mazurek et al., 2026 [PMID 42685447]).
- **Nature-based activities:** Parent-reported benefits in communication/calmness (Galbraith et al., 2026 [PMID 42603141]).
- **Oral hygiene:** Peer modeling interventions **under investigation** (NCT07859696).

Key studies

2. **Mazurek et al. (2026):** Systematic review (39 studies) showed **behavioral sleep interventions** significantly improved sleep outcomes (Mazurek et al., *Sleep Medicine*).
3. ****Restoy et al.**
4. **Galbraith et al. (2026):** Qualitative study (n=15) highlighted **parent-reported benefits** of nature walks for communication/calmness (Galbraith et al., *International Journal of Qualitative Studies on Health and Well-being*).

Conflicting or negative evidence

None retrieved for direct contradictions or negative results in ASD treatment. Terminated trial (NCT05307744) cited **operational challenges** (COVID-19, recruitment slowdown) but did not report efficacy data.

What remains under investigation

- **Music therapy dosage:** Systematic review protocol (Chen et al., 2026 [PMID 42486527]) awaits results.
- **Oral hygiene:** Peer modeling trials (NCT07859696) not yet recruiting.
- **Rhythm/gait:** Completed trial (NCT06659328) on structured walking **results not posted**.
- **Genetic/environmental targets:** No direct treatment implications from twin studies (Han et al., 2026 [PMID 42508582]). (*indirect evidence: different or related population*)

Treatment Drug Landscape

Regulatory status (official FDA drug labels)

Drug (generic / brand)	Category	Approved indication (FDA label wording)	Dose / route	Approval status / date	Source
Aripiprazole	FDA-approved for the labelled indication	"1 INDICATIONS AND USAGE Aripiprazole is indicated for the treatment of: • Schizophrenia • Irritability Associated with Autistic Disorder • Treatment of Tourette's Disorder Aripiprazole is an atypical antipsychotic. The oral formulations ..."	Not retrieved (see FDA label)	FDA label effective 2026-03-05; original approval date not retrieved	[DailyMed label](https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=02a4af27-c83c-4166-950c-7a1cb12d198d)
Risperidone	FDA-approved for the labelled indication	"with lithium or valproate, for the treatment of acute manic or mixed episodes associated with Bipolar I Disorder (1.2) Treatment of irritability associated with autistic disorder (1.3) 1.1 Schizophrenia Risperidone tablets are ..."	Not retrieved (see FDA label)	FDA label effective 2026-06-25; original approval date not retrieved	[DailyMed label](https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=03d3b52d-3a7d-4595-84c9-332da6ca9171)

Approval applies only to the labelled indication quoted above; it is not an approval for other symptoms or features of the condition. Dose, route and frequency are not extracted here: see the linked FDA label.

Other agents named in the retrieved evidence (not in the FDA label records)

None retrieved.

Safety alerts: None retrieved.

Clinical trials (registry data)

Treatment and intervention trials

Trial	Intervention	Phase	Registry status (last update)	Reading
NCT05307744	PS128; Placebo	NA	TERMINATED (2026-10-06)	Terminated early (During COVID-19, study execution and participant compliance were ...); results posted (not retrieved here).
NCT06659328	Structured Walking Intervention for Children with Autism Spectrum Disorder.	NA	COMPLETED (2026-10-06)	Completed; results not posted/retrieved.
NCT07858383	Behavioral: PRIME-ASD (Parent-led Intervention Module for Early Autism Spectrum ...	NA	NOT_YET_RECRUITING (2026-10-05)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT04903353	Comparison of Risperidone and Aripiprazole	PHASE4	COMPLETED (2026-10-05)	Completed; results not posted/retrieved.
NCT07375043	Group Drumming	NA	COMPLETED (2026-10-05)	Completed; results not posted/retrieved.
NCT07859345	Leap Motion-Based Virtual Reality Intervention	NA	NOT_YET_RECRUITING (2026-10-05)	Under investigation; results not yet available; effectiveness cannot be inferred.

Other registry records (assessment, characterisation or peripheral research; not treatment trials)

Trial	Intervention	Phase	Registry status (last update)	Reading
NCT03718923	FOXP1 Syndrome: The Seaver Autism Center for Research and Treatment is Characterizing ...	NA	RECRUITING (2026-10-06)	Under investigation; results not yet available; effectiveness cannot be inferred. Specific syndrome record; do not generalise to the whole condition.
NCT07859696	Peer model; Pre-recorded video of peer modeling	NA	NOT_YET_RECRUITING (2026-10-06)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07858617	assessment	NA	NOT_YET_RECRUITING (2026-10-05)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06933342	Habituation to dental examination; Preparations at home	NA	RECRUITING (2026-10-05)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06391684	Social/non social; predictable vs unpredictable	NA	RECRUITING (2026-10-02)	Under investigation; results not yet available; effectiveness cannot be inferred.

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Evidence limitations

- Findings are drawn from abstracts and web pages, not full texts; study quality was not appraised from the full papers.
- Doses and approval dates appear only where a retrieved source states them; otherwise they are marked 'Not retrieved'.
- Excluded retracted publication: Efficacy of oral folinic acid supplementation in children with autism spectrum disorder: a (PMID 39243316, 2024). Retracted work is not used as evidence.
- 2 retrieved record(s) were excluded because they were not about the question's subject.
- Source `pubmed\_fast\_q`: retrieval partial (timed out after 3s (partial)); evidence status: unavailable. The absence of evidence cannot be concluded from this source.
- Source `fda\_fast`: retrieval partial (timed out after 3s (partial)); evidence status: unavailable. The absence of evidence cannot be concluded from this source.
- Source `web\_fast`: retrieval partial (timed out after 3s (partial)); evidence status: unavailable. The absence of evidence cannot be concluded from this source.

Sources

- Galbraith C et al. Walking together: exploring australian parent experiences of walking in nature with their preschool children with autism using a bioecological lens. *International journal of qualitative studies on health and well-being*. 2026 Dec 31. PMID 42603141. DOI 10.1080/17482631.2026.2719072
- Chen Y et al. Effect of music therapy dosage on social communication and emotion regulation in children with autism spectrum disorder: study protocol for a systematic review and meta-analysis. *BMJ open*. 2026-07-22. PMID 42486527. DOI 10.1136/bmjopen-2026-119412
- Fisberg M et al. FEEDSETUP: feeding difficulties spectrum evaluation and treatment up-to-date proposal. *Jornal de pediatria*. 2026-09-18. PMID 42716484. DOI 10.1016/j.jpmed.2026.101610

4. Chadha M et al. AllPlay Dance autism: protocol for a randomised controlled trial of a community-based dance programme for children with autism. *BMJ open*. 2026-09-08. PMID 42711081. DOI 10.1136/bmjopen-2026-122285
5. Mazurek MO et al. Behavioral interventions for sleep problems in autistic children: A systematic review. *Sleep medicine*. 2026 Dec. PMID 42685447. DOI 10.1016/j.sleep.2026.109242
6. Wozniak J et al. A randomized trial of omega-3 fatty acids plus inositol versus N-acetylcysteine for the treatment of depression and mania in emotionally dysregulated youth age 5-17 with and without autism traits. *Journal of affective disorders*. 2026 Dec 1. PMID 42462928. DOI 10.1016/j.jad.2026.122277
7. Kemp E et al. AquOTic: protocol for a type 1 hybrid effectiveness-implementation trial of an occupational therapy-based water competency intervention for children on the autism spectrum. *BMJ open*. 2026-08-04. PMID 42552003. DOI 10.1136/bmjopen-2026-117908
8. Han Y et al. Genetic and environmental origins of the co-occurrence of autistic traits, anxiety and depression: Evidence from meta-analysis and twin study. *Journal of affective disorders*. 2026 Dec 1. PMID 42508582. DOI 10.1016/j.jad.2026.122317
9. Wang T et al. Dose-response relationship between exercise interventions and motor skills in children with autism spectrum disorder: a systematic review and multilevel meta-analysis. *European journal of pediatrics*. 2026-09-18. PMID 42758334. DOI 10.1007/s00431-026-07421-4
10. Floríndez LI et al. Oral care interventions for autistic individuals: A systematic review. *Autism : the international journal of research and practice*. 2025 Jan. PMID 39056277. DOI 10.1177/13623613241260171
11. Restoy D et al. Emotion regulation and emotion dysregulation in children and adolescents with Autism Spectrum Disorder: A meta-analysis of evaluation and intervention studies. *Clinical psychology review*. 2024 Apr. PMID 38401510. DOI 10.1016/j.cpr.2024.102410
12. FDA drug label: Aripiprazole (label effective 2026-03-05).
<https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=02a4af27-c83c-4166-950c-7a1cb12d198d>
13. FDA drug label: Risperidone (label effective 2026-06-25).
<https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=03d3b52d-3a7d-4595-84c9-332da6ca9171>

Verification (run on this final text): PASS - 12 PMIDs, 11 DOIs, 11 trial IDs and 2 links checked against retrieved data. Repair passes: 1. Drug landscape status: complete.

- Repaired: removed 1 line(s): NCT07076264 is not in the retrieved registry data
- Repaired: removed sentence with unsupported value 'n=583'
- Repaired: removed sentence with unsupported value 'n=1,245'
- Repaired: removed unsupported claim: the claim mentions 'social communication', which the cited source (PMID 42758334) does not report
- Repaired: removed protocol-as-efficacy claim (PMID 42552003)
- Repaired: softened 'confirmed/proven/validated' (a single study cannot 'confirm' or 'prove') (x2)
- Repaired: reworded dose-response language as an exploratory finding (x2)
- Repaired: labelled the cited paper as indirect evidence (different or related population) (x2)
- Repaired: relabelled 'Established' as 'Evidence-supported but limited' (no guideline among the cited sources)